Project Goal

The **Berka dataset**, from the **1999 PKDD Discovery Challenge**, provides information on the clients, accounts, and transactions of a Czech bank. The original task description of the Discovery Challenge states:

The bank wants to improve their services. For instance, the bank managers have only vague idea, who is a good client (whom to offer some additional services) and who is a bad client (whom to watch carefully to minimize the bank loses). Fortunately, the bank stores data about their clients, the accounts (transactions within several months), the loans already granted, the credit cards issued The bank managers hope to improve their understanding of customers and seek specific actions to improve services. A mere application of a discovery tool will not be convincing for them.

In keeping with the original task description, our project goal is to mine and analyze this bank data in order to extrapolate from it the type of customer who makes a good candidate for a credit card.

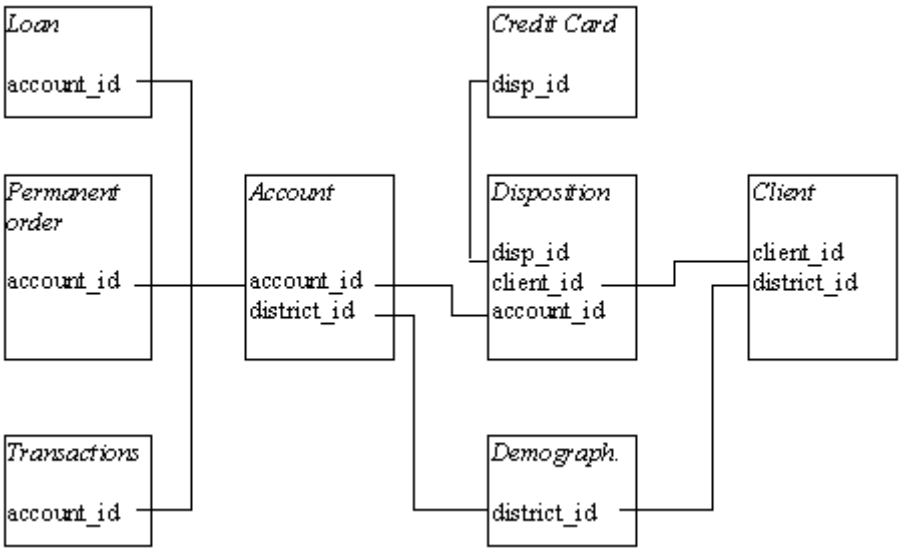
Domain Description

This database was prepared by Petr Berka and Marta Sochorova.
We will refer to the dataset as *The Berka Dataset* throughout this report.

The Berka dataset is a collection of financial information from a Czech bank. The dataset deals with over 5,300 bank clients with approximately 1,000,000 transactions. Additionally, the bank represented in the dataset has extended close to 700 loans and issued nearly 900 credit cards, all of which are represented in the data.

Data Description

Entity-Relationship Description



- Each account has both static characteristics (e.g. date of creation, address of the branch) given in relation "account" and dynamic characteristics (e.g. payments debited or credited, balances) given in relations "permanent order" and "transaction".
- Relation "client" describes characteristics of persons who can manipulate with the accounts.
- One client can have more accounts, more clients can manipulate with single account; clients and accounts are related together in relation "disposition".
- Relations "loan" and "credit card" describe some services which the bank offers to its clients;
- More than one credit card can be issued to an account,
- At most one loan can be granted for an account.
- Relation "demographic data" gives some publicly available information about the districts (e.g. the unemployment rate); additional information about the clients can be deduced from this.

Table Descriptions

ACCOUNTS
Each record describes static characteristics of an account
Size: 4500 objects in the file
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CLIENTS
Each record describes characteristics of a client
Size : 5369 objects in the file
Return To Top

DISPOSITON (DISP)
Each record relates together a client with an account i.e. this relation describes the rights of clients to operate accounts
Size: 5369 objects in the file
Return To Top

PERMANENT ORDERS, Debits only (ORDER)
Each record describes characteristics of a payment order
Size : 6471 objects in the file
Return To Top

TRANSACTIONS (TRANS)
Each record describes one transaction on an account
Size: 1056320 objects in the file
Return To Top

LOANS
Each record describes a loan granted for a given account
Size: 682 objects in the file
Return To Top

CREDIT CARDS (CARD)
Each record describes a credit card issued to an account
Size : 892 objects in the file
Return To Top

DEMOGRAPHIC DATA (DISTRICT)
Each record describes demographic characteristics of a district
Size: 77 objects in the file
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RESULTS:

The following rules represent our findings in analyzing whether or not a customer will/will not have a credit card

1. If the average account balance is less-than-or-equal 37097.18, then the client will not be a card holder.
2. If the client is middle-age and lives in Prague, and his/her average account balance is greater-than 66899.77, then that client will be a card holder.
3. If the client lives in Prostejov, and has an average account balance in excess of 46123.76, then that client will be a card holder.

The following are characterizations of cardholders within the current population of bank clients

1. There is no apparent difference between men and women in the issuance of credit cards
2. Middle aged single females have a higher precentage of classic and gold cards than the general population.
3. All credit card customers that have taken a loan have good status in repayment.
4. Customers who are in the age bracket less than 24 and whose average balance is under 50000.00 are likely candidates for a JUNIOR Card.
5. Otherwise, it is not easily determined from customer information which card type has been selected. Perhaps further examination of customer behavior is warranted.